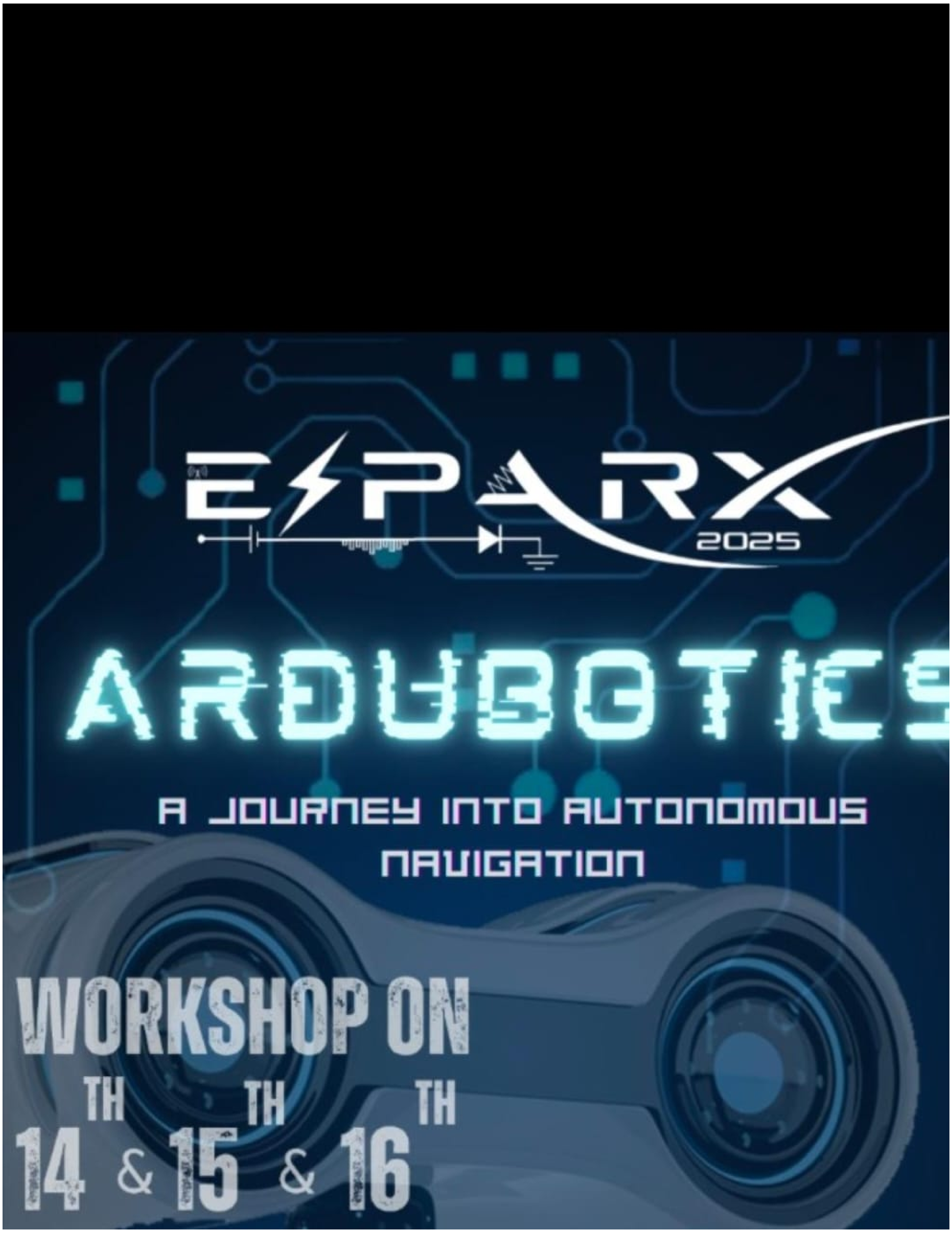

Line Follower Robot

Overview:

This project involves designing and building an autonomous Line Follower Robot that can detect and follow a predefined path using infrared (IR) sensors. The robot is programmed to move along a black line on a white surface by continuously sensing the line’s position and adjusting its direction accordingly. The system uses Arduino as the microcontroller, IR sensors for line detection. The logic enables the robot to make real-time decisions, such as turning left, right, or moving straight, based on sensor input.

⚙Components Used:

1. Arduino UNO
2. IR Sensor Module (Left & Right)
3. L293D Motor Driver Module
4. DC Motors & Wheels
5. Robot Chassis
6. Jumper Wires
7. Battery Pack
8. USB Cable

Features Of Line Follower Robot:

1. **Automatic Path Tracking** – Follows a black line on a white surface without human control.

2. **Cost-Effective Project** – Built using easily available and affordable components.

3. **Real-Time Sensor Response** – IR sensors continuously detect the line and guide movement instantly.

Code:

```
#define enA 10

#define in1 9
```

```
#define in2 8
#define in3 7
#define in4 6
#define enB 5
#define left_IR A0
#define right_IR A1

void setup() {
  Serial.begin(9600);

  pinMode(enA, OUTPUT);
  pinMode(in1, OUTPUT);
  pinMode(in2, OUTPUT);
  pinMode(in3, OUTPUT);
  pinMode(in4, OUTPUT);
  pinMode(enB, OUTPUT);
  pinMode(left_IR, INPUT);
  pinMode(right_IR, INPUT);

  analogWrite(enA, 120);   // Set motor A speed
  analogWrite(enB, 140);   // Set motor B speed
}

void loop() {
```

```
  int left_value = digitalRead(left_IR);
  int right_value = digitalRead(right_IR);

  if (left_value == 1 && right_value == 0) {
    right();   // Turn right
  }
  else if (right_value == 1 && left_value == 0) {
    left();    // Turn left
  }
  else if (right_value == 0 && left_value == 0) {
    forward(); // Move forward
  }
  else if (right_value == 1 && left_value == 1) {
    STOP();    // Stop at dead end
  }
  else {
    STOP();    // Default safety stop
  }
}

void right() {
  digitalWrite(in1, LOW);
  digitalWrite(in2, HIGH);
  digitalWrite(in3, LOW);
```

```
  digitalWrite(in4, HIGH);
}

void left() {
  digitalWrite(in1, HIGH);
  digitalWrite(in2, LOW);
  digitalWrite(in3, HIGH);
  digitalWrite(in4, LOW);
}

void STOP() {
  digitalWrite(in1, LOW);
  digitalWrite(in2, LOW);
  digitalWrite(in3, LOW);
  digitalWrite(in4, LOW);
}

void forward() {
  digitalWrite(in1, HIGH);
  digitalWrite(in2, LOW);
  digitalWrite(in3, LOW);
  digitalWrite(in4, HIGH);
}
```

Working Video Demonstration Link:

https://drive.google.com/file/d/1BxZruryFcUkGIVs1HWlaIWi5AZhiyKXj/view?usp=drivesdk